

Fixing Boston's Busiest Equity Corridors: Routes 23, 28 & 66

20% Faster Commutes.

1.23M Rider Hours Saved Annually.

2 Year Payback.

TartanSparks

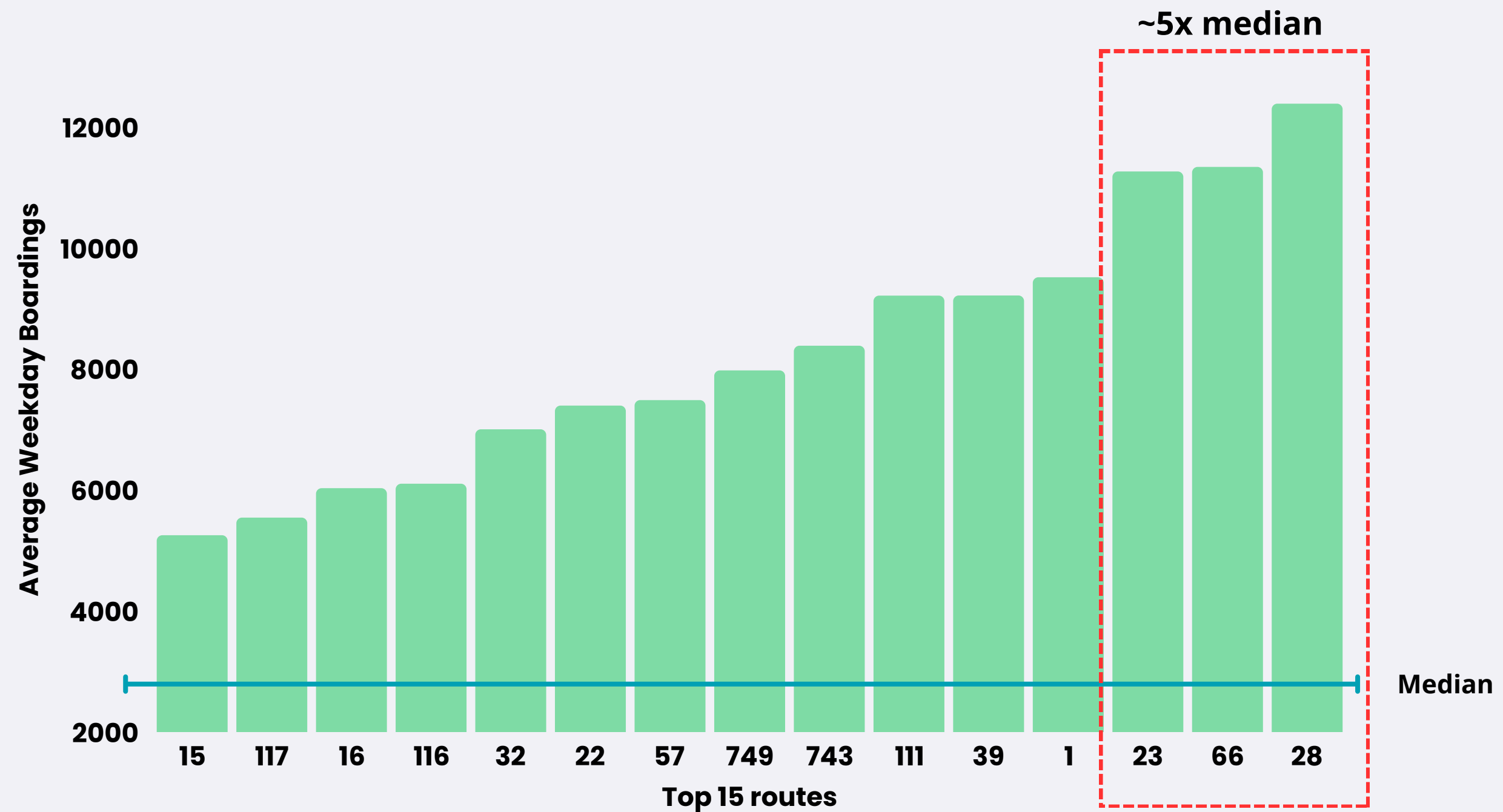
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November 23, 2025



Why These Corridors Matter

MBTA bus ridership has **rebounded faster than rail since covid**. Routes 23, 28, 66 each carry **10–12+ weekday boardings**, far above the bus system median (~2,400)



Riders on these routes disproportionately live in **zero-car, low-income, majority-non-white households** yet they experience **slower trips and less reliable service** than system averages

How We Identified Equity Corridors

Step 1: Transit Dependence Index – **Who needs the bus most?**

Built a Transit Dependence Index (TDI) at the census-tract level using four components:

- % households with **no vehicle**
- % population **below 200%** of the federal poverty line
- % **non-white** residents
- % **workers** who commute by transit

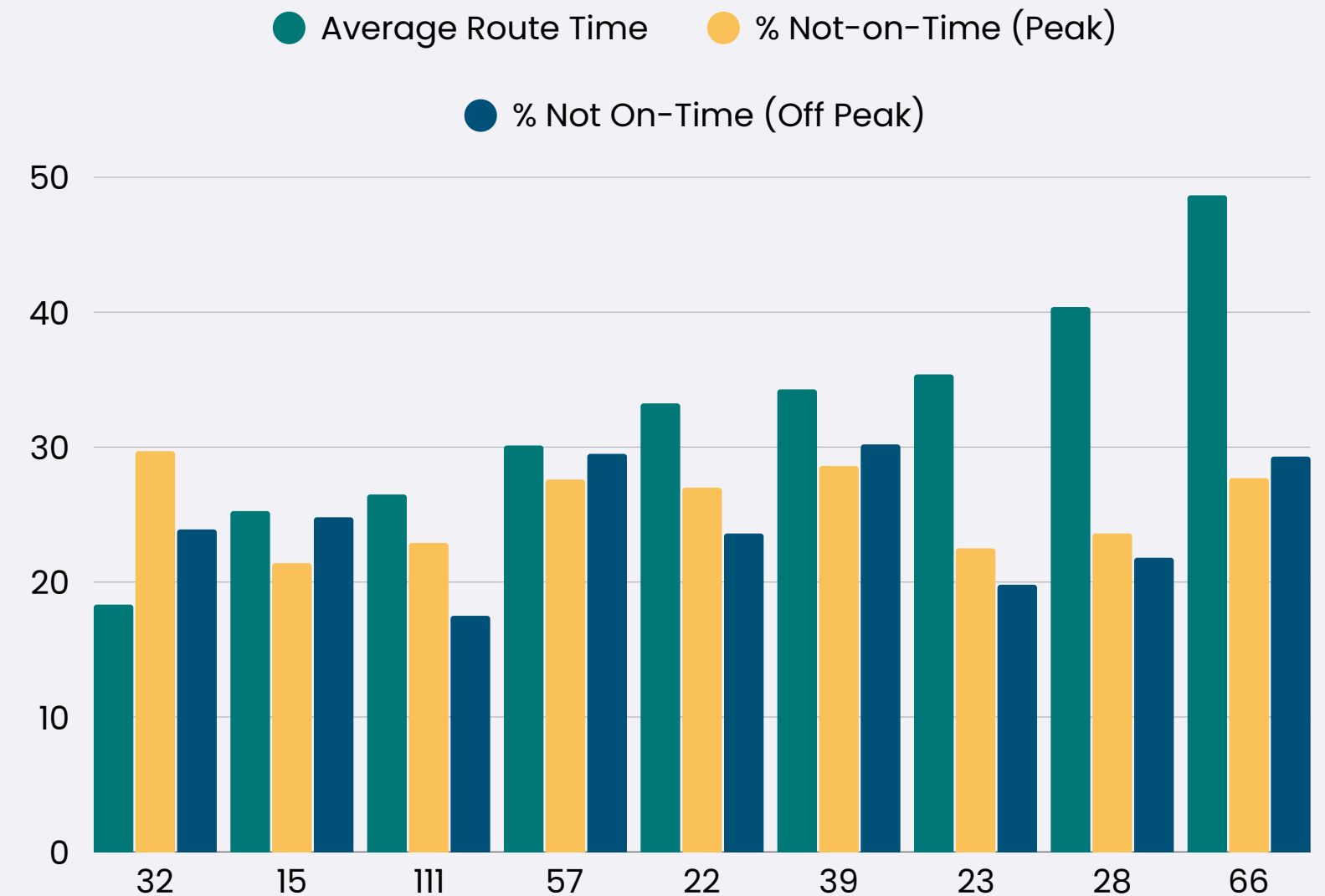
Normalized each factor to system averages, then averaged into a composite TDI

Ranked all major bus routes by TDI × ridership (Equity volume score)

Routes 23, 28, and 66 are the top three equity-volume corridors in the entire bus network yet



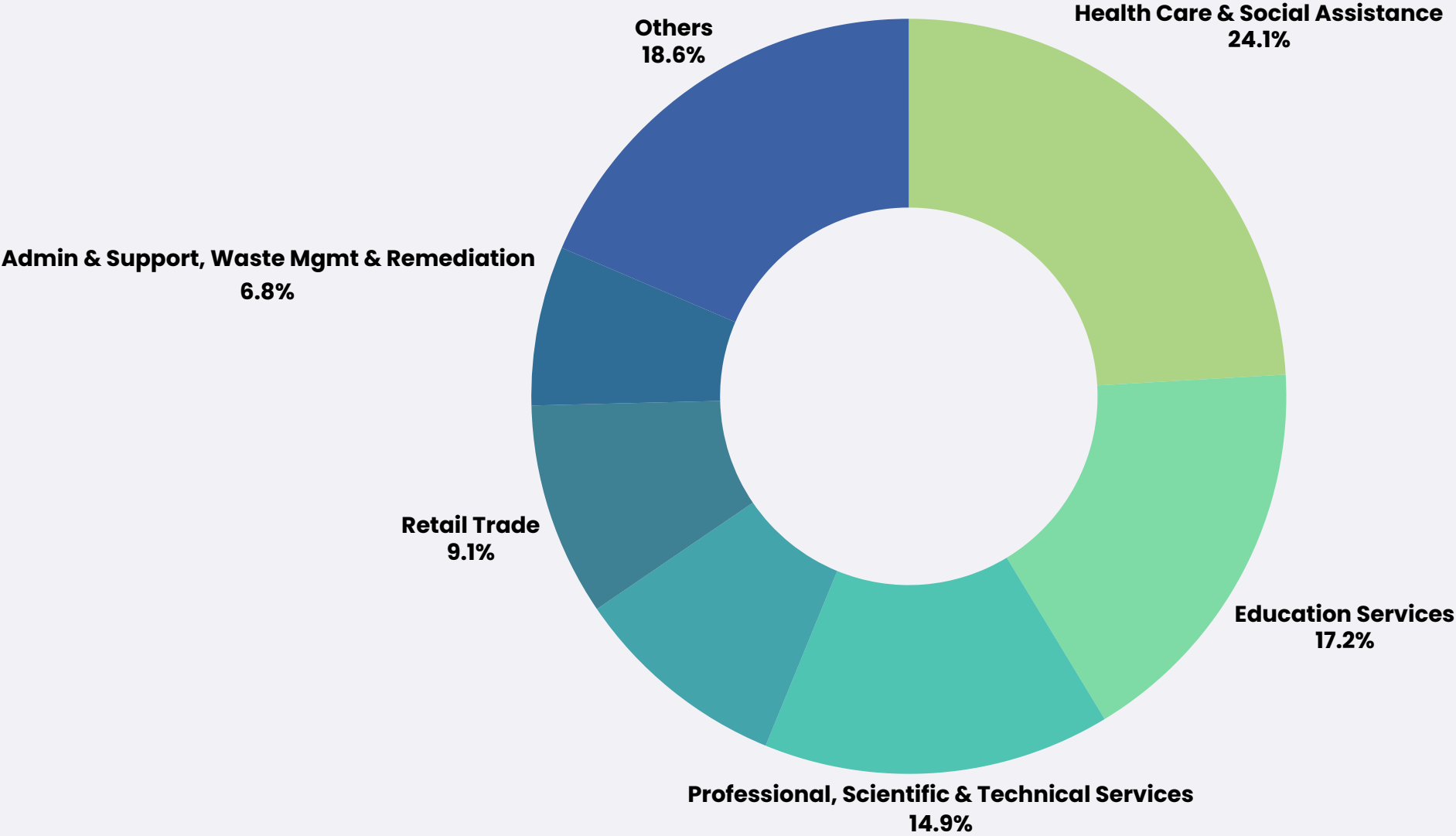
Step 2: Service Deficit – **Who gets the slowest, least reliable trips?**



All three routes have longer-than-median peak runtimes. Route 66 is especially slow.

Busy but slow: reliability gaps on key corridors

Low Wages, Long Trips, Critical Roles



Main sectors of jobs held by ENG corridor residents (top 10)

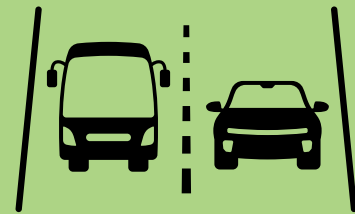
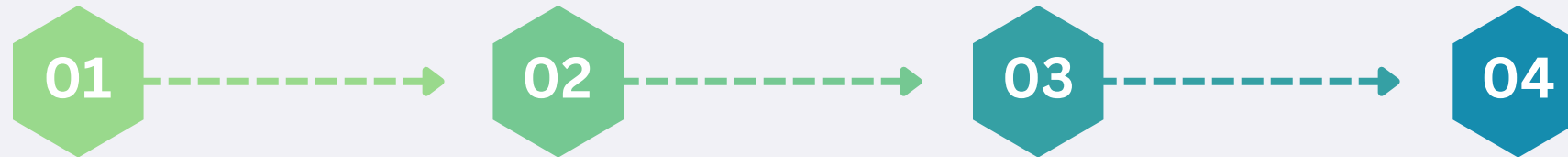
Type*	Low Wage Job Share	Mid wage Job share	High Wage Job Share
ENG residents	13.4	21.5	65.1
Rest MA residents	12.3	19.7	68.1

Only about **one-quarter of jobs** held by **ENG residents** are within the corridor. The vast majority of workers **must travel beyond the corridor**, often using rail via bus connections, to reach employment.

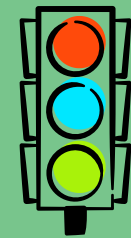
The ENG corridor is a **major regional job center** anchored by health care and education, sectors that depend heavily on frequent, reliable, on-time transit for both workers and residents.

Better bus speed and reliability within and through this corridor expand local job access while improving regional commuting efficiency and job connectivity far beyond the corridor itself.

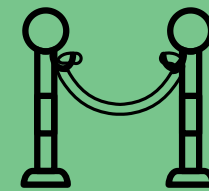
A Proven Bus-Priority Toolkit, Not a Mega-Project



**Dedicated
bus lanes at
congested
segment**



**Transit Signal
Priority at key
intersections**



**Queue jumps
at
chokepoints**



**Stop
consolidation
& far-side
stops**

Enforce existing parking & bus lane rules – Mounted Cameras



Dedicated lanes, signal priority, queue jumps, and smarter stops on key segments can deliver **~15– 20% faster trips using tools Boston already deploys.**

Fiscal Case: Costs, Savings, Payback [1/2]

The ENG corridor improvements would include:

- Dedicated / priority bus lanes on:
 - Blue Hill Avenue / Warren Street segments for routes 23 & 28
 - High-delay segments of route 66 (e.g., Harvard Ave, Brookline–Roxbury corridor)
- Transit Signal Priority (TSP) at key intersections along all three routes.
- Queue jumps at identified bottlenecks.
- Stop consolidation & far-side stops where warranted.
- Off-board payment / all-door boarding infrastructure for Route 66.

Potential funding sources include:



Federal

FTA CMAQ (Congestion Mitigation and Air Quality) grants for bus priority and mode shift.
FTA Small Starts or Bus and Bus Facilities funds.



State

MassDOT and MBTA capital programs focused on state of good repair + climate.
State bonds targeting emissions reductions and workforce access.



City of Boston

Capital funds under the Boston Transportation Department, building on prior bus projects.



Value capture / institutional contributions

Institutional partners (major hospitals and universities along the ENG corridor) that benefit from better workforce access could contribute via shuttle consolidation savings or voluntary contributions.

Total: ~10 miles of priority treatment

23 & 28: ~**6 mi** of overlapping or adjacent segments needing treatment

66: ~**4 mi** of high-delay segments

Base cost = 10 mi * \$0.5M/mi = \$5.0M
Total with overhead ≈ 1.3 × 5.0M = \$6.5M

Capital program
≈ **\$6–7 million**

Fiscal Case: Costs, Savings, Payback [2/2]

Direct Savings

20% faster → 20% fewer hours needed for the same timetable:

$$\text{Annual savings} = 85.1 \times 297 \times 250 = \$6.32\text{M}$$

Hours saved/day Opex per bus hour Weekdays

If we assume half of that gets reinvested into

Better frequencies | More schedule padding | Recovery time | Extension of priority to evenings/weekends

Then:

Net Annual savings ≈ \$3.16M/year

Payback: ~1–2 years; The 10-year net present value of direct MBTA **financial benefits is approximately \$23 million**, against a one-time capital cost of \$6.5 million.

Alternate Revenue Benefit:

When bus speeds improve and ridership rises (~15%), **roughly 25% of those extra bus trips are likely to become new rail transfers.**

At a conservative \$1.70 average rail fare, that's about \$2,200 per weekday, **~\$0.56 million per year in new rail revenue.**

Indirect Savings

Route 23

36.04 min

Route 28

41.10 min

Route 66

48.94 min

-20%

**7.21 min
saved per
trip**

**8.22 min
saved per
trip**

**9.79 min
saved per
trip**

Annually,

~1.23M rider-hours saved

At \$10/hour → ≈ \$12.3M/year

At \$20/hour → ≈ \$24.5M/year

\$12–24M

in rider time value

plus job access and equity effects

Implementation and Risk Mitigation

Implementation Roadmap

Phase 1 (Year 1): Design + Quick-Build on 23/28

Focus on Blue Hill Ave / Warren St.
Implement continuous bus lanes where feasible, TSP at top delay intersections, and stop optimization.

Phase 2 (Year 2): Extend to Route 66

Identify 3–4 highest delay segments (e.g., around Harvard Ave, Brookline Village, Roxbury Crossing).

Combine bus lanes, queue jumps, and off-board payment.

Phase 3 (Year 3+): Job-access targeting and refinement

Use refined GTFS + LODES analysis to prioritize any remaining gaps in access to major health and education employers and to escalate to more capital-intensive upgrades where justified.

Risks & mitigations

Local opposition to lane reallocation/parking loss

Use before-and-after data from similar Boston projects showing improved travel times and minimal impact on local business; provide loading zones and side-street parking management.

MBTA capacity constraints (operators, mechanics)

Emphasize that ENG reduces operator hours for the same service or allows more service with the same workforce, aligning with the agency's staffing constraints.

Uncertain ridership rebound

ENG corridors focus on already high-ridership routes with robust transit dependence, limiting downside risk.

Future considerations

- Explore a **permanent fare-free policy** on route 66, given that only a small share of corridor revenue is actually generated through fares.
- Extend the proposed bus-priority package to **other high-ridership corridors** such as Routes 1, 117, and 39, etc.



THANK YOU

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